

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Scenario Based Assessment)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A

Scenario Based Assessment

Code of Conduct:

I Mandati Sharan Mahi Reddy (192324082) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Mandati Sharan Mahi Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

ANNEXURE

Q1: Student Registration System

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

A)

Three Collection Interfaces

- List
- Set
- Map

B)

Representation

HashSet<Student>

|
v

HashMap<Student, Course>

C) Java Program

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and the user's name (MANDATI SHARAN MAHI REDDY) with a dropdown arrow. The main content area is titled "Q1 Student Registration System" and contains the problem description and questions. A sidebar on the left lists five questions, with the first one selected. The "Your Code" section shows a Java program that uses HashSet and HashMap to store student data and their enrolled courses. The "Input" section has a text area for stdin input, and the "Output" section shows the expected output for the given input.

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class StudentRegistration {
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<>();
5         students.add("Teja");
6         students.add("Ravi");
7         students.add("Anil");
8         HashMap<String, String> courses = new HashMap<>();
9         courses.put("Teja", "Java");
10        courses.put("Ravi", "Python");
11        courses.put("Anil", "AI");
12        Iterator<String> it = students.iterator();
13        while (it.hasNext()) {
14            String name = it.next();
15            System.out.println(name + " -> " + courses.get(name));
16        }
17    }
18 }
```

Input

stdin input

Output

```
Ravi -> Python
Teja -> Java
Anil -> AI
```

5/5 answered

Run Save

Q2: E-Commerce Cart System

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
- Add items
 - Remove an item
 - Display all items

A)

Three Collection Classes

- ArrayList
- LinkedList
- Vector

B)

Representation

ArrayList

Laptop

Mouse

Keyboard

Phone

C) Java Program

The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and a user profile for MANDATI SHARAN MAHI REDDY. The main content area is titled 'E-Commerce Cart System' and contains a question about implementing a shopping cart using ArrayList and Iterator. A 'Your Code' section shows a Java program that adds items to a cart and displays them. The 'Output' section shows the result of the program execution.

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         ArrayList<String> cart = new ArrayList<>();
5         cart.add("Laptop");
6         cart.add("Mouse");
7         cart.add("Keyboard");
8         String item = "Mouse";
9         ArrayList<String> newCart = new ArrayList<>();
10        Iterator<String> it = cart.iterator();
11        while (it.hasNext()) {
12            String x = it.next();
13            if (!x.equals(item)) {
14                newCart.add(x);
15            }
16        }
17        System.out.println("Items in Cart:");
18        Iterator<String> it2 = newCart.iterator();
19        while (it2.hasNext()) {
20            System.out.println(it2.next());
21        }
22    }
23 }
```

Input

stdin input

Output

Items in Cart:
Laptop
Keyboard

Run Save

Q3: Library Management System

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

A)

Three Map Implementations

- HashMap
- TreeMap
- LinkedHashMap

B)

Representation

ISBN	Book
------	------

101	Java
102	Python
103	AI

C) Java Program

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

MANDATI SHARAN MAHI REDDY
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CO2-AT1-Scenario Based Programs

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QUESTIONS**Q1**
Student Registration System
10 marks**Q2**
E-Commerce Cart System
10 marks**Q3**
Library Management System
10 marks**Q4**
Employee Data Processing
10 marks**Q5**
Online Voting System
10 marks

5/5 answered

Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```
1 import java.util.*;
2 public class Library {
3     public static void main(String[] args) {
4         HashMap<Integer, String> books = new HashMap<>();
5         books.put(101, "Java");
6         books.put(102, "Python");
7         books.put(103, "AI");
8         System.out.println("Book with ISBN 102 : " + books.get(102));
9         for (Integer key : books.keySet()) {
10             System.out.println(key + " -> " + books.get(key));
11         }
12     }
13 }
```

Input

stdin input

OutputBook with ISBN 102 : Python
101 -> Java
102 -> Python
103 -> AI

Run Save

Q4: Employee Data Processing (10 Marks)

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

A)

Advantages of Generics

- Type Safety
- Code Reusability
- No Type Casting

B)

Representation

List<Employee>

Employee

Name

Salary

C) Java Program

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the SIMATS Engineering logo, the lab name, and a user profile. The main content area is titled 'CO2-AT1-Scenario Based Programs' and contains a list of questions on the left and a detailed view of question Q4 on the right. Q4 is 'Employee Data Processing' (10 marks). The question text is: 'Suppose that a company processes employee records and filters employees based on salary. (a) List three advantages of Generics in Collections. (b) Represent how employee data is stored using List with Generics. (c) Write a Java program using Iterator to: • Store employee objects • Display employees with salary > 50,000'. The 'Your Code' section shows a Java program that defines an Employee class with name and salary attributes, and a main method that creates a list of employees, adds them, and iterates through them to print those with a salary greater than 50,000. The 'Input' section is empty, and the 'Output' section shows the expected output: 'Bob - 60000.0' and 'Charlie - 55000.0'.

SIMATS Online Programming Lab

MANDATI SHARAN MAHI REDDY 192324052RA

CO2-AT1-Scenario Based Programs

Q4 Employee Data Processing 10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 public class Employee {
5     String name;
6     double salary;
7     public Employee(String name, double salary) {
8         this.name = name;
9         this.salary = salary;
10    }
11    public static void main(String[] args) {
12        List<Employee> employees = new ArrayList<>();
13        employees.add(new Employee("Alice", 45000));
14        employees.add(new Employee("Bob", 60000));
15        employees.add(new Employee("Charlie", 55000));
16        employees.add(new Employee("David", 48000));
17        Iterator<Employee> it = employees.iterator();
18        while (it.hasNext()) {
19            Employee emp = it.next();
20            if (emp.salary > 50000) {
21                System.out.println(emp.name + " - " + emp.salary);
22            }
23        }
24    }
25 }
```

Input

stdin input

Output

```
Bob - 60000.0
Charlie - 55000.0
```

Q5: Online Voting System (10 Marks)

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results

A)

Three Collection Types

- HashSet
- HashMap
- List

B)

Representation

HashSet -> Voter IDs

101

102

103

HashMap

Candidate	Votes
-----------	-------

A	2
---	---

B	1
---	---

C) Java Program

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

MANDATI SHARAN MAHI REDDY
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CO2-AT1-Scenario Based Programs← Back

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```
1 import java.util.*;
2 public class VotingSystem {
3     public static void main(String[] args) {
4         HashSet<Integer> voters = new HashSet<>();
5         voters.add(101);
6         voters.add(102);
7         voters.add(101);
8         HashMap<String, Integer> votes = new HashMap<>();
9         votes.put("A", 2);
10        votes.put("B", 1);
11        System.out.println("Voters:");
12        for (Integer id : voters) {
13            System.out.println(id);
14        }
15        System.out.println("Results:");
16        for (String c : votes.keySet()) {
17            System.out.println(c + " -> " + votes.get(c));
18        }
19    }
20 }
```

Run Save

Input

stdin input

Output

```
Voters:
101
102
Results:
A -> 2
B -> 1
```